

LOAD BALANCING IN AWS

❖ Why Use a Load Balancer?

- "A load balancer is used to **distribute incoming** traffic across multiple servers to ensure better performance, high availability, and reliability of applications."

➤ Key Reasons to Use a Load Balancer:

- 1. Distributes Traffic Evenly**
 - Prevents one server from getting overloaded.
- 2. Improves Application Availability**
 - If one server fails, traffic is redirected to healthy servers.
- 3. Enhances Performance**
 - Balances the load so users get faster responses.
- 4. Supports Auto Scaling**
 - Works well with auto scaling groups to handle changing traffic.
- 5. Enables Fault Tolerance**
 - Keeps your application running even if one or more servers go down.
- 6. Secure Traffic Management**
 - Can handle SSL termination, hide internal server IPs, and filter requests.
- 7. Allows Rolling Updates with Zero Downtime**
 - Send traffic to new servers while updating the old ones.

➤ Types of Load Balancers in AWS

AWS offers three types of Load Balancers under the Elastic Load Balancing (ELB) service:

1. *Application Load Balancer (ALB)*
2. *Network Load Balancer (NLB)*
3. *Gateway Load Balancer (GWLB)*

1. Application Load Balancer (ALB)

- **Layer:** Works at **Layer 7** (Application Layer)
- **Best for:** HTTP, HTTPS, and gRPC traffic
- **Features:**
 - Content-based routing (URL path, host-based)
 - Supports WebSocket
 - Target groups by IP, instance, or Lambda
 - Native integration with **containers (ECS)**

2. Network Load Balancer (NLB)

- **Layer:** Works at **Layer 4** (Transport Layer)
- **Best for:** TCP, UDP, TLS traffic with ultra-low latency
- **Features:**
 - Handles millions of requests per second
 - Supports static IPs and Elastic IPs
 - Ideal for high-performance or latency-sensitive applications

3. Gateway Load Balancer (GWLB)

- **Layer:** Works at **Layer 3 (Network Layer)**
- **Best for:** Deploying **third-party virtual appliances** (e.g., firewalls, IDS/IPS)
- **Features:**
 - Transparent traffic inspection

- Easy to insert appliances inline without changing routing

✕ Difference Between Types of Load Balancers in AWS

Feature	ALB	NLB	GWLB
OSI Layer	Layer 7 – Application	Layer 4 – Transport	Layer 3 – Network
Protocol Support	HTTP, HTTPS, gRPC	TCP, UDP, TLS	Any IP traffic
Routing Type	Content-based (host/path) routing	Connection-based	Packet-based forwarding
Target Types	EC2, IP, Lambda	EC2, IP	Third-party appliances (via GENEVE)
Static IP Support	✕ No	✓ Yes	✓ Yes
Health Checks	HTTP/HTTPS based	TCP/HTTP	VPC-level appliance health

[General Interview Questions]

◆ 1. What is a Load Balancer in AWS and why is it used?

- A Load Balancer distributes incoming network traffic across multiple targets (like EC2 instances) to ensure high availability, fault tolerance, and scalability.

◆ 2. How does an ALB route traffic differently than NLB?

- ALB uses host-based and path-based routing (e.g., /images to one group, /api to another).
- NLB routes based on IP and port only

◆ 3. What is a Target Group in ELB?

- A logical group of targets (EC2, IPs, or Lambda) that receive traffic from a Load Balancer.

◆ 4. Can a single ALB route traffic to multiple microservices? How?

- Yes, by configuring **path-based routing** or **host-based routing** rules linked to different target groups.

◆ 5. What is the benefit of NLB supporting static IPs?

- It allows services needing **fixed IPs** (e.g., firewall rules, whitelisting) to **function without DNS** dependency.

◆ 6. How does an ELB perform health checks?

- It pings the target on a configured **path/port** (like /health), and **removes unhealthy** targets from routing.

◆ 7. What happens if all targets in a target group fail health checks?

- "If all targets in a target group fail health checks, the load balancer will stop sending traffic to them. As a result, users may see errors or timeouts because there are no healthy instances to handle the requests."

🔍 Explanation:

- The **Load Balancer only routes traffic to healthy targets.**
- If **all are unhealthy**, it has **nowhere to send traffic.**

◆ 11. How do you secure Load Balancer endpoints?

- Use HTTPS listeners
- Attach SSL/TLS certificates
- Use WAF (Web Application Firewall)
- Restrict access via security groups/NACLs

◆ 12. What are listeners in ELB?

- A **listener** is a process that checks for incoming connection requests on configured **ports** and **protocols**, and forwards them to target groups.